RESEARCH LETTER

Racism in Schools and Health Among US Adolescents in the United States, 2023

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Introduction

Racism is one of the best documented and most powerful social determinants of health.[1,2] However, there has been relatively little research studying the role that racism plays in the health of children.[3] In this study, we examine the relationship between experiences with racism at school and important health behaviors among a nationally representative sample of US high school students.

Methods

This study uses data from the Centers for Disease Control and Prevention's 2023 Youth Risk Behavior Surveillance System (YRBS; $N = 20,103$; overall response rate 35.4%), a biannual survey administered in US high schools on behavioral health topics.[4] The survey uses a three-stage cluster design to capture a representative sample of high school students. The survey includes measures of mental health, substance use, and violence victimization, among other important health behaviors (eTable 1 in the Supplement). In 2023, the national YRBS introduced a measure of children's experience with racism at school. Children were asked "How often have you felt that you were treated badly or unfairly in school because of your race or ethnicity?" with possible responses "Never," "Rarely," "Sometimes," "Most of the time," and "Always."

First, the unadjusted prevalence of several risky behaviors is reported by level of experience with racism. Then, the relative risk of each of these behaviors was estimated via survey-weighted logistic regression, accounting for sex, age, race/ethnicity, English language proficiency, and grades, relative to those who experienced no racism. These analyses were conducted collectively and by student race/ethnicity. Estimates are presented as risk ratios, which are computed using random draws from the regression's variance-covariance matrix and holding confounders at their means.[5] All analyses account for the YRBS complex survey design to produce nationally representative estimates and correct standard errors.

This study is exempt from institutional review as it only uses secondary analysis of publicly

available data.

Results

Of the 18,286 YRBS participants with non-missing data, 31.5% (95% CI, 29.0%–34.1%) reported experiencing some racism at school, including 56.9% of Asian, 45.9% of Black, 42.5% of Multiracial, 36.4% of Hispanic, and 17.3% of White children (Table 1). 5.7% of non-White children reported experiencing racism at school most of the time or always.

Most negative health behaviors exhibit a dose-response relationship with children's experience with racism. For example, suicidal ideation has a prevalence of 16.8% among children who report never experiencing racism, 25.6% among children who report experiencing racism rarely or sometimes, and 42.7% among children who report experiencing racism most of the time or always.

These patterns persist after accounting for potential confounders (Figure 1). After controlling for sex, age, race/ethnicity, English language proficiency, and grades, those who experience racism rarely or sometimes (RR 2.26, 95% CI 1.87–2.67) and those who experience racism most of the time or always (RR 4.42, 95% CI 3.64–5.33) are significantly more likely to be bullied at school compared with those who never experience racism. The response is strongest for Asian children (rarely or sometimes RR 3.81, 95% CI 1.23–11.61; most of the time or always RR 26.75, 95% CI 8.15–73.83) and Black children (rarely or sometimes RR 2.87, 95% CI 1.97–4.36; most of the time or always RR 6.61, 95% CI 3.92–10.62) (eTable 2 in the Supplement).

Discussion

A significant proportion of US high school children report experiencing racism in schools, and these experiences correlate significantly and in a dose-response relationship with several negative health outcomes.

Racism is a well-recognized threat to public health.[1,2] These results extend that finding to

children in schools[3] and merit a well-coordinated response from public health professionals. Recent studies find that a composite measure of racial mistreatment and school connection is strongly correlated with substance use and mental health behaviors among American adolescents.[6,7] The present study complements these findings by examining a broader set of health outcomes in a larger, nationally representative sample. Policy-makers, educators, and clinicians may design interventions to reduce racism among young people and to screen children who experience racism at school for negative health behaviors.[8]

This study has several limitations. This cross-sectional study cannot establish causation. The YRBS is designed to be nationally representative, but its 35.4% response rate could limit generalizability. Though children respond privately and anonymously, self-report can be biased. The survey uses multiple choice responses, which precludes more detailed accounts of children's experience with racism. The survey did not capture student experiences with racism outside of school.

Future research should seek a clearer understanding of how children experience racism at school, the causal effects of racism on child health, and which policies and interventions can reduce children's experience with racism.

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Table 1: Weighted Prevalence (%) of Health Outcomes by Frequency of School-Based Racial Discrimination, 2023 YRBS

Panel A: Full Sample ($N = 18,286$)

Outcome	N	Weighted Prevalence, % (95% CI)				
		Never ($n = 12,530$)	Rarely ($n = 2,836$)	Sometimes ($n = 2,024$)	Most of the time ($n = 471$)	Always ($n = 425$)
Mental Health						
Persistent sadness or hopelessness	18,094	33.8 (31.8–35.7)	48.8 (45.4–52.2)	58.1 (54.9–61.2)	68.8 (62.5–75.0)	62.2 (55.6–68.8)
Seriously considered suicide	18,025	16.8 (15.2–18.3)	22.6 (20.1–25.2)	30.0 (26.5–33.6)	43.7 (36.2–51.1)	41.7 (32.8–50.6)
Made a suicide plan	16,789	13.4 (11.7–15.1)	19.8 (16.1–23.4)	23.6 (19.4–27.8)	39.3 (30.5–48.2)	34.1 (25.7–42.4)
Attempted suicide	17,590	6.7 (5.8–7.7)	11.0 (9.0–13.1)	15.5 (13.3–17.6)	26.1 (18.1–34.1)	33.0 (24.1–41.9)
Violence						
Bullied at school	18,169	14.8 (13.0–16.6)	20.2 (17.0–23.3)	32.3 (27.4–37.1)	47.5 (39.5–55.6)	56.6 (48.9–64.3)
Electronically bullied	18,141	12.8 (11.2–14.5)	18.0 (13.9–22.0)	24.9 (20.7–29.1)	39.7 (28.5–50.9)	45.1 (36.1–54.2)
Physical fight at school	17,176	5.9 (5.1–6.7)	9.1 (7.5–10.8)	12.5 (10.4–14.6)	17.2 (11.6–22.7)	35.8 (29.0–42.5)
Carried weapon at school	17,813	3.1 (2.1–4.0)	4.0 (2.8–5.3)	5.8 (3.6–8.1)	12.5 (7.4–17.5)	18.8 (13.3–24.3)
Substance Use						
Current cigarette use	17,974	2.9 (2.2–3.7)	3.5 (2.1–4.8)	3.1 (1.7–4.5)	7.4 (3.7–11.2)	18.0 (11.3–24.7)
Current e-cigarette use	17,432	14.7 (13.2–16.1)	18.4 (16.0–20.9)	18.8 (15.6–22.0)	28.5 (22.1–34.9)	35.4 (27.7–43.2)
Current alcohol use	17,486	20.0 (18.1–22.0)	25.8 (23.0–28.7)	24.1 (20.5–27.6)	29.9 (23.6–36.1)	41.7 (32.4–51.0)
Current marijuana use	17,843	14.5 (13.1–16.0)	19.5 (16.5–22.5)	21.5 (17.6–25.5)	29.9 (21.3–38.4)	33.5 (24.9–42.2)

Panel B: Non-White Students ($N = 9,312$)

Outcome	N	Weighted Prevalence, % (95% CI)				
		Never ($n = 5,185$)	Rarely ($n = 1,926$)	Sometimes ($n = 1,547$)	Most of the time ($n = 349$)	Always ($n = 305$)
Mental Health						
Persistent sadness or hopelessness	9,196	30.7 (28.5–33.0)	48.8 (44.8–52.8)	57.2 (53.4–61.0)	67.3 (59.7–74.9)	63.4 (56.8–69.9)
Seriously considered suicide	9,151	12.7 (11.4–14.1)	21.3 (18.7–23.8)	27.9 (24.1–31.8)	43.1 (35.1–51.1)	41.9 (29.9–53.9)
Made a suicide plan	8,953	11.5 (9.8–13.1)	17.7 (15.0–20.4)	22.5 (18.1–26.9)	40.5 (33.4–47.5)	30.2 (21.4–39.0)
Attempted suicide	8,904	6.3 (5.1–7.6)	11.0 (8.8–13.3)	16.3 (13.9–18.7)	28.6 (20.3–37.0)	34.0 (23.8–44.1)
Violence						
Bullied at school	9,248	8.7 (7.0–10.5)	16.5 (13.0–20.0)	27.7 (23.0–32.5)	45.0 (36.4–53.6)	55.3 (46.5–64.1)
Electronically bullied	9,225	8.2 (6.5–9.9)	14.7 (11.3–18.1)	20.5 (16.5–24.6)	37.0 (27.5–46.5)	41.5 (31.2–51.9)
Physical fight at school	8,627	6.7 (5.2–8.2)	9.0 (6.9–11.2)	12.2 (9.9–14.5)	16.8 (10.4–23.2)	35.7 (28.3–43.1)
Carried weapon at school	8,938	2.3 (1.6–3.0)	3.3 (2.2–4.4)	5.1 (3.1–7.2)	10.0 (4.9–15.2)	16.0 (9.6–22.5)
Substance Use						
Current cigarette use	9,119	1.4 (0.8–2.0)	2.1 (0.9–3.2)	1.7 (0.7–2.7)	6.4 (2.4–10.5)	17.0 (10.5–23.6)
Current e-cigarette use	8,798	11.8 (10.2–13.3)	17.0 (14.5–19.6)	17.5 (14.4–20.7)	30.3 (23.1–37.5)	39.0 (29.5–48.5)
Current alcohol use	8,857	14.9 (13.3–16.5)	23.9 (20.8–27.1)	21.1 (17.6–24.7)	32.4 (25.0–39.7)	40.7 (28.6–52.7)
Current marijuana use	9,034	12.9 (11.4–14.4)	19.0 (15.7–22.2)	20.8 (17.1–24.5)	34.0 (23.5–44.5)	37.5 (25.4–49.6)

Note. CI = confidence interval. N = unweighted analytic sample. n = unweighted students at each level of school-based racial discrimination. Prevalences are weighted to the YRBS complex three-stage cluster design.

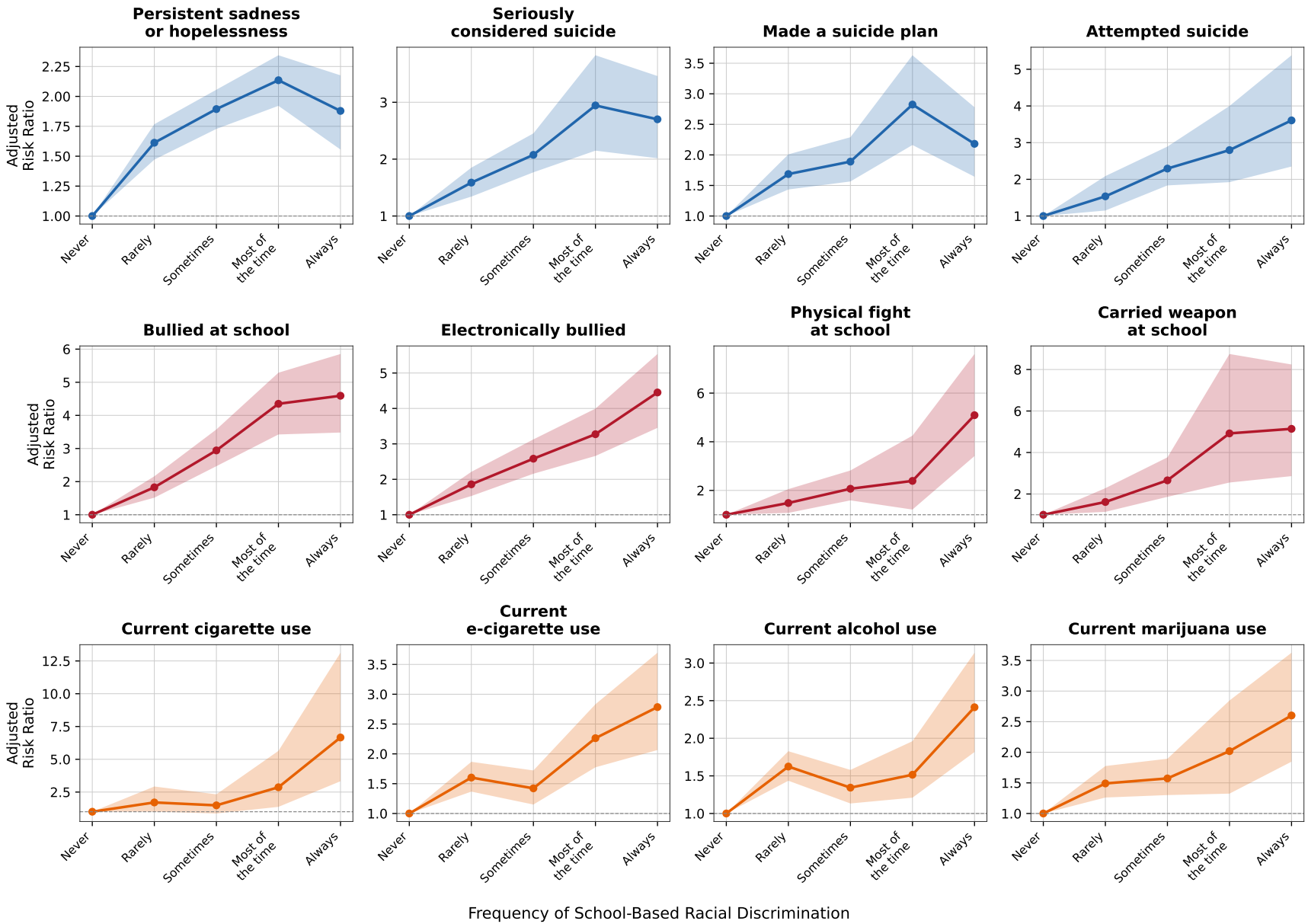


Figure 1: Adjusted risk ratios for health outcomes by frequency of school-based racism among US high school students, 2023 YRBS ($N = 18,286$). Risk ratios from logistic regression computed via simulation from the variance-covariance matrix (King et al, 2000), adjusting for sex, age, race/ethnicity, English language proficiency, and grades. Shaded bands indicate 95% confidence intervals. Reference category is Never.

Supplement

The 2023 National YRBS had an overall response rate of 35.4% (school response rate 49.8% × student response rate 71.0%).[4]

eTable 1. Survey Items and Analytic Coding, 2023 National Youth Risk Behavior Survey

Variable	Survey Question	Response Options / Coding
Exposure		
School-based racism (Q23)	"During your life, how often have you felt that you were treated badly or unfairly in school because of your race or ethnicity?"	1 = Never; 2 = Rarely; 3 = Sometimes; 4 = Most of the time; 5 = Always. Coded as categorical (5 levels; reference = Never).
Outcomes — Mental Health		
Persistent sadness (QN26)	"During the past 12 months, did you ever feel so sad or hopeless almost every day for two weeks or more in a row that you stopped doing some usual activities?"	Yes / No. Coded as binary.
Suicidal ideation (QN27)	"During the past 12 months, did you ever seriously consider attempting suicide?"	Yes / No. Coded as binary.
Suicide plan (QN28)	"During the past 12 months, did you make a plan about how you would attempt suicide?"	Yes / No. Coded as binary.
Suicide attempt (QN29)	"During the past 12 months, how many times did you actually attempt suicide?"	0 times / 1 time / 2–3 / 4–5 / 6+. Coded as binary (1 if ≥1 time).
Outcomes — Violence		
Bullied at school (QN24)	"During the past 12 months, have you ever been bullied on school property?"	Yes / No. Coded as binary.
Electronically bullied (QN25)	"During the past 12 months, have you ever been electronically bullied (counting being bullied through texting, Instagram, Facebook, or other social media)?"	Yes / No. Coded as binary.
Physical fight at school (QN17)	"During the past 12 months, how many times were you in a physical fight on school property?"	0 / 1 / 2–3 / 4–5 / 6–7 / 8–9 / 10–11 / 12+ times. Coded as binary (1 if ≥1 time).
Carried weapon at school (QN12)	"During the past 30 days, on how many days did you carry a weapon such as a gun, knife, or club on school property?"	0 / 1 / 2–3 / 4–5 / 6+ days. Coded as binary (1 if ≥1 day).
Outcomes — Substance Use		

Variable	Survey Question	Response Options / Coding
Current cigarette use (QN33)	"During the past 30 days, on how many days did you smoke cigarettes?"	0 / 1–2 / 3–5 / 6–9 / 10–19 / 20–29 / All 30 days. Coded as binary (1 if ≥1 day).
Current e-cigarette use (QN36)	"During the past 30 days, on how many days did you use an electronic vapor product?"	0 / 1–2 / 3–5 / 6–9 / 10–19 / 20–29 / All 30 days. Coded as binary (1 if ≥1 day).
Current alcohol use (QN42)	"During the past 30 days, on how many days did you have at least one drink of alcohol?"	0 / 1–2 / 3–5 / 6–9 / 10–19 / 20–29 / All 30 days. Coded as binary (1 if ≥1 day).
Current marijuana use (QN48)	"During the past 30 days, how many times did you use marijuana?"	0 / 1–2 / 3–9 / 10–19 / 20–39 / 40+ times. Coded as binary (1 if ≥1 time).
Covariates		
Sex (Q2)	"What is your sex?"	Female / Male. Coded as binary (1 = Female).
Age (Q1)	"How old are you?"	12 years or younger / 13 / 14 / 15 / 16 / 17 / 18 or older. Coded as categorical (5 levels: ≤14, 15, 16, 17, 18+; ages ≤14 collapsed because of sparse counts at 12 and 13).
Race/ethnicity (raceeth)	Derived from Q4 ("Are you Hispanic or Latino?") and Q5 ("What is your race?").	8 categories: AI/AN, Asian, Black, NH/PI, White, Hispanic, Multiple-Hispanic, Multiple-Non-Hispanic. Coded as categorical (8 levels).
Grades (QN87)	"During the past 12 months, how would you describe your grades in school?"	Mostly A's / B's / C's / D's / F's / None / Not sure. Coded as binary (1 if A's or B's).
English proficiency (QN107)	"How well do you speak English?"	Very well / Well / Not well / Not at all. Coded as binary (1 if Very well or Well).

eTable 2. Adjusted Risk Ratios for Health Outcomes by Frequency of School-Based Racial Discrimination, Stratified by Race/Ethnicity, 2023 YRBS

All Students ($N = 18,286$)

		Adjusted Risk Ratio (95% CI)				
Outcome	N	Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	10,299	1 [Ref]	1.61 (1.47–1.77)	1.89 (1.73–2.06)	2.14 (1.92–2.34)	1.88 (1.56–2.18)
Seriously considered suicide	10,263	1 [Ref]	1.59 (1.34–1.85)	2.08 (1.77–2.45)	2.94 (2.15–3.83)	2.70 (2.02–3.46)
Made a suicide plan	10,270	1 [Ref]	1.69 (1.43–2.01)	1.89 (1.57–2.29)	2.82 (2.16–3.63)	2.18 (1.64–2.78)
Attempted suicide	10,284	1 [Ref]	1.54 (1.15–2.09)	2.29 (1.83–2.89)	2.80 (1.92–4.00)	3.61 (2.35–5.38)
Violence						
Bullied at school	10,318	1 [Ref]	1.83 (1.51–2.16)	2.94 (2.47–3.58)	4.35 (3.42–5.29)	4.59 (3.48–5.85)
Electronically bullied	10,312	1 [Ref]	1.86 (1.53–2.21)	2.58 (2.16–3.13)	3.27 (2.66–4.00)	4.45 (3.45–5.54)
Physical fight at school	10,294	1 [Ref]	1.48 (1.08–2.05)	2.07 (1.59–2.82)	2.39 (1.21–4.25)	5.09 (3.41–7.61)
Carried weapon at school	10,334	1 [Ref]	1.61 (1.14–2.28)	2.66 (1.86–3.77)	4.92 (2.55–8.75)	5.14 (2.86–8.24)
Substance Use						
Current cigarette use	10,315	1 [Ref]	1.71 (1.01–2.92)	1.49 (0.87–2.33)	2.87 (1.37–5.64)	6.66 (3.33–13.13)
Current e-cigarette use	9,949	1 [Ref]	1.60 (1.37–1.87)	1.42 (1.15–1.72)	2.26 (1.77–2.83)	2.78 (2.06–3.70)
Current alcohol use	10,195	1 [Ref]	1.62 (1.44–1.83)	1.34 (1.13–1.58)	1.52 (1.21–1.96)	2.41 (1.82–3.14)
Current marijuana use	10,270	1 [Ref]	1.49 (1.26–1.78)	1.57 (1.30–1.89)	2.02 (1.32–2.85)	2.60 (1.85–3.63)

Non-White Students ($N = 9,312$)

		Adjusted Risk Ratio (95% CI)				
Outcome	N	Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	5,797	1 [Ref]	1.75 (1.57–1.95)	1.96 (1.75–2.17)	2.19 (1.88–2.52)	2.03 (1.68–2.39)
Seriously considered suicide	5,775	1 [Ref]	1.81 (1.50–2.19)	2.28 (1.83–2.84)	3.45 (2.35–4.86)	3.29 (2.34–4.34)
Made a suicide plan	5,781	1 [Ref]	1.80 (1.52–2.18)	1.98 (1.58–2.46)	3.49 (2.48–4.71)	2.10 (1.54–2.80)
Attempted suicide	5,795	1 [Ref]	1.69 (1.21–2.32)	2.74 (2.05–3.60)	3.45 (2.11–5.60)	4.70 (3.27–6.79)
Violence						
Bullied at school	5,817	1 [Ref]	2.33 (1.77–3.05)	3.33 (2.50–4.32)	5.85 (4.36–7.68)	6.33 (4.74–8.69)
Electronically bullied	5,813	1 [Ref]	2.19 (1.72–2.78)	2.73 (2.12–3.50)	3.93 (2.85–5.28)	5.60 (3.76–7.90)
Physical fight at school	5,795	1 [Ref]	1.56 (0.99–2.38)	2.00 (1.37–2.79)	1.98 (0.93–3.84)	4.64 (3.24–6.98)
Carried weapon at school	5,823	1 [Ref]	1.67 (0.87–3.20)	3.28 (1.95–5.63)	6.07 (3.11–11.50)	7.12 (3.78–12.67)
Substance Use						
Current cigarette use	5,819	1 [Ref]	1.85 (0.82–4.68)	1.65 (0.73–3.72)	3.30 (1.26–9.03)	10.28 (5.03–19.92)
Current e-cigarette use	5,572	1 [Ref]	1.74 (1.39–2.17)	1.45 (1.10–1.87)	2.47 (1.77–3.40)	3.40 (2.47–4.56)
Current alcohol use	5,734	1 [Ref]	1.81 (1.53–2.09)	1.36 (1.09–1.68)	1.73 (1.26–2.35)	3.05 (2.05–4.19)
Current marijuana use	5,784	1 [Ref]	1.56 (1.25–1.93)	1.65 (1.32–2.12)	2.31 (1.44–3.49)	3.30 (2.08–4.64)

White Students ($N = 8,733$)

Outcome	N	Adjusted Risk Ratio (95% CI)				
		Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	4,502	1 [Ref]	1.39 (1.21–1.59)	1.85 (1.51–2.16)	2.11 (1.55–2.55)	1.55 (0.78–2.20)
Seriously considered suicide	4,488	1 [Ref]	1.31 (1.00–1.72)	1.92 (1.59–2.29)	2.37 (1.43–3.34)	1.74 (1.10–2.61)
Made a suicide plan	4,489	1 [Ref]	1.54 (1.08–2.15)	1.75 (1.29–2.36)	1.43 (0.67–2.64)	2.43 (1.30–4.14)
Attempted suicide	4,489	1 [Ref]	1.52 (0.93–2.52)	1.53 (0.87–2.75)	1.59 (0.87–2.87)	1.22 (0.34–4.16)
Violence						
Bullied at school	4,501	1 [Ref]	1.35 (1.05–1.74)	2.72 (2.21–3.31)	3.01 (1.87–3.99)	2.72 (1.11–4.27)
Electronically bullied	4,499	1 [Ref]	1.49 (1.11–2.00)	2.56 (1.88–3.37)	2.75 (1.76–3.77)	2.98 (1.60–4.51)
Physical fight at school	4,499	1 [Ref]	1.22 (0.74–2.08)	2.40 (1.45–3.94)	3.74 (1.27–9.46)	6.66 (2.79–12.55)
Carried weapon at school	4,511	1 [Ref]	1.65 (1.13–2.34)	1.94 (1.23–3.07)	4.77 (2.15–9.50)	3.09 (1.30–6.84)
Substance Use						
Current cigarette use	4,496	1 [Ref]	1.64 (0.99–2.84)	1.40 (0.68–2.83)	2.16 (0.63–6.81)	2.85 (0.65–10.11)
Current e-cigarette use	4,377	1 [Ref]	1.40 (1.06–1.80)	1.46 (1.03–2.06)	1.75 (1.07–2.83)	1.64 (0.59–3.63)
Current alcohol use	4,461	1 [Ref]	1.36 (1.06–1.71)	1.34 (1.06–1.69)	1.08 (0.60–1.68)	1.60 (0.83–2.67)
Current marijuana use	4,486	1 [Ref]	1.38 (1.03–1.81)	1.48 (1.05–2.02)	1.39 (0.86–2.17)	1.29 (0.35–3.40)

Black Students ($N = 1,732$)

Outcome	N	Adjusted Risk Ratio (95% CI)				
		Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	1,220	1 [Ref]	1.62 (1.24–2.08)	1.87 (1.48–2.35)	2.43 (1.74–2.99)	1.72 (1.10–2.47)
Seriously considered suicide	1,215	1 [Ref]	1.63 (1.11–2.39)	2.07 (1.35–3.12)	3.12 (1.25–6.06)	1.62 (0.70–3.19)
Made a suicide plan	1,221	1 [Ref]	1.79 (1.33–2.39)	1.75 (1.13–2.76)	2.60 (0.80–6.45)	2.45 (0.90–5.19)
Attempted suicide	1,220	1 [Ref]	2.90 (1.70–5.04)	3.02 (1.79–5.43)	4.52 (1.28–12.28)	2.80 (0.89–7.76)
Violence						
Bullied at school	1,221	1 [Ref]	2.71 (1.61–4.30)	3.10 (2.00–4.89)	8.08 (4.17–13.33)	5.83 (3.03–10.02)
Electronically bullied	1,219	1 [Ref]	2.86 (1.87–4.42)	1.91 (1.06–3.47)	6.51 (2.51–13.21)	3.90 (1.64–8.51)
Physical fight at school	1,212	1 [Ref]	2.01 (1.13–3.50)	1.08 (0.63–1.77)	2.81 (0.76–7.01)	4.20 (1.90–7.65)
Carried weapon at school	1,222	1 [Ref]	0.57 (0.17–1.73)	1.13 (0.42–3.06)	12.40 (2.73–37.51)	4.72 (1.59–12.35)
Substance Use						
Current cigarette use	1,225	1 [Ref]	1.67 (0.21–14.15)	0.87 (0.14–5.06)	1.41 (0.08–23.15)	6.68 (1.99–19.39)
Current e-cigarette use	1,163	1 [Ref]	1.67 (1.01–2.73)	1.71 (1.15–2.47)	2.51 (0.65–6.72)	4.15 (2.75–5.95)
Current alcohol use	1,204	1 [Ref]	2.13 (1.30–3.37)	1.40 (0.90–2.01)	2.16 (0.85–4.60)	3.35 (1.49–5.91)
Current marijuana use	1,221	1 [Ref]	1.81 (1.23–2.60)	1.90 (1.39–2.67)	2.95 (1.22–5.40)	2.86 (1.63–4.63)

Hispanic Students ($N = 1,179$)

		Adjusted Risk Ratio (95% CI)				
Outcome	N	Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	569	1 [Ref]	1.96 (1.39–2.89)	2.29 (1.72–3.16)	3.43 (2.63–4.58)	1.07 (0.26–2.53)
Seriously considered suicide	561	1 [Ref]	2.32 (1.18–4.55)	3.05 (2.04–4.49)	8.68 (1.73–16.37)	2.60 (0.31–9.83)
Made a suicide plan	568	1 [Ref]	3.10 (1.96–4.88)	4.61 (2.63–7.76)	9.49 (2.53–16.59)	0.00 (0.00–0.00)
Attempted suicide	563	1 [Ref]	1.31 (0.61–2.78)	3.13 (1.40–7.38)	5.46 (1.63–12.94)	2.41 (0.16–14.26)
Violence						
Bullied at school	571	1 [Ref]	1.35 (0.73–2.52)	2.02 (1.00–4.09)	3.72 (0.66–10.19)	3.54 (0.76–11.54)
Electronically bullied	569	1 [Ref]	2.28 (0.84–5.68)	2.53 (0.98–5.89)	8.15 (1.43–21.88)	1.94 (0.17–11.52)
Physical fight at school	573	1 [Ref]	1.00 (0.34–3.17)	3.05 (1.23–7.19)	0.89 (0.12–4.78)	0.00 (0.00–0.00)
Carried weapon at school	569	1 [Ref]	10.85 (3.07–35.82)	11.26 (2.14–60.62)	0.94 (0.09–10.12)	0.00 (0.00–0.00)
Substance Use						
Current cigarette use	572	1 [Ref]	0.00 (0.00–0.00)	0.04 (0.00–0.46)	—	0.00 (0.00–0.00)
Current e-cigarette use	552	1 [Ref]	2.20 (1.28–3.70)	0.89 (0.45–1.66)	2.93 (0.57–8.85)	1.53 (0.17–6.94)
Current alcohol use	566	1 [Ref]	2.54 (1.55–4.26)	2.17 (1.26–3.61)	1.81 (0.52–4.64)	2.58 (0.38–8.79)
Current marijuana use	568	1 [Ref]	2.12 (1.07–3.96)	0.35 (0.11–1.25)	2.33 (0.39–9.54)	3.62 (0.69–12.75)

Asian Students ($N = 927$)

Outcome	N	Adjusted Risk Ratio (95% CI)				
		Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	396	1 [Ref]	2.44 (1.66–3.53)	3.35 (2.28–4.79)	4.25 (2.34–6.67)	5.40 (2.88–7.82)
Seriously considered suicide	393	1 [Ref]	4.08 (1.77–11.04)	6.97 (2.08–23.84)	12.80 (3.05–45.10)	13.39 (3.26–48.98)
Made a suicide plan	392	1 [Ref]	3.23 (1.12–11.81)	1.47 (0.52–4.71)	10.71 (3.40–31.84)	2.57 (0.31–15.24)
Attempted suicide	397	1 [Ref]	1.21 (0.31–4.58)	3.15 (1.25–7.56)	16.65 (5.26–49.32)	6.65 (1.04–29.62)
Violence						
Bullied at school	394	1 [Ref]	1.79 (0.43–6.83)	8.55 (2.42–28.47)	20.19 (4.99–64.42)	—
Electronically bullied	396	1 [Ref]	2.23 (0.71–6.43)	5.09 (1.49–15.78)	1.35 (0.27–6.65)	12.34 (2.74–36.51)
Physical fight at school	398	1 [Ref]	1.07 (0.23–4.74)	4.36 (0.48–45.21)	—	0.62 (0.02–13.01)
Carried weapon at school	398	1 [Ref]	0.78 (0.04–13.45)	15.89 (2.01–99.18)	—	0.00 (0.00–0.00)
Substance Use						
Current cigarette use	396	1 [Ref]	—	—	—	—
Current e-cigarette use	389	1 [Ref]	2.69 (0.66–10.51)	6.08 (0.66–48.74)	—	1.82 (0.13–20.66)
Current alcohol use	399	1 [Ref]	0.94 (0.36–2.32)	2.43 (1.28–4.43)	2.65 (0.27–11.58)	2.72 (0.31–10.48)
Current marijuana use	395	1 [Ref]	0.76 (0.13–3.99)	3.51 (0.78–17.58)	2.70 (0.35–14.60)	0.14 (0.01–1.93)

American Indian/Alaska Native Students ($N = 1,086$)

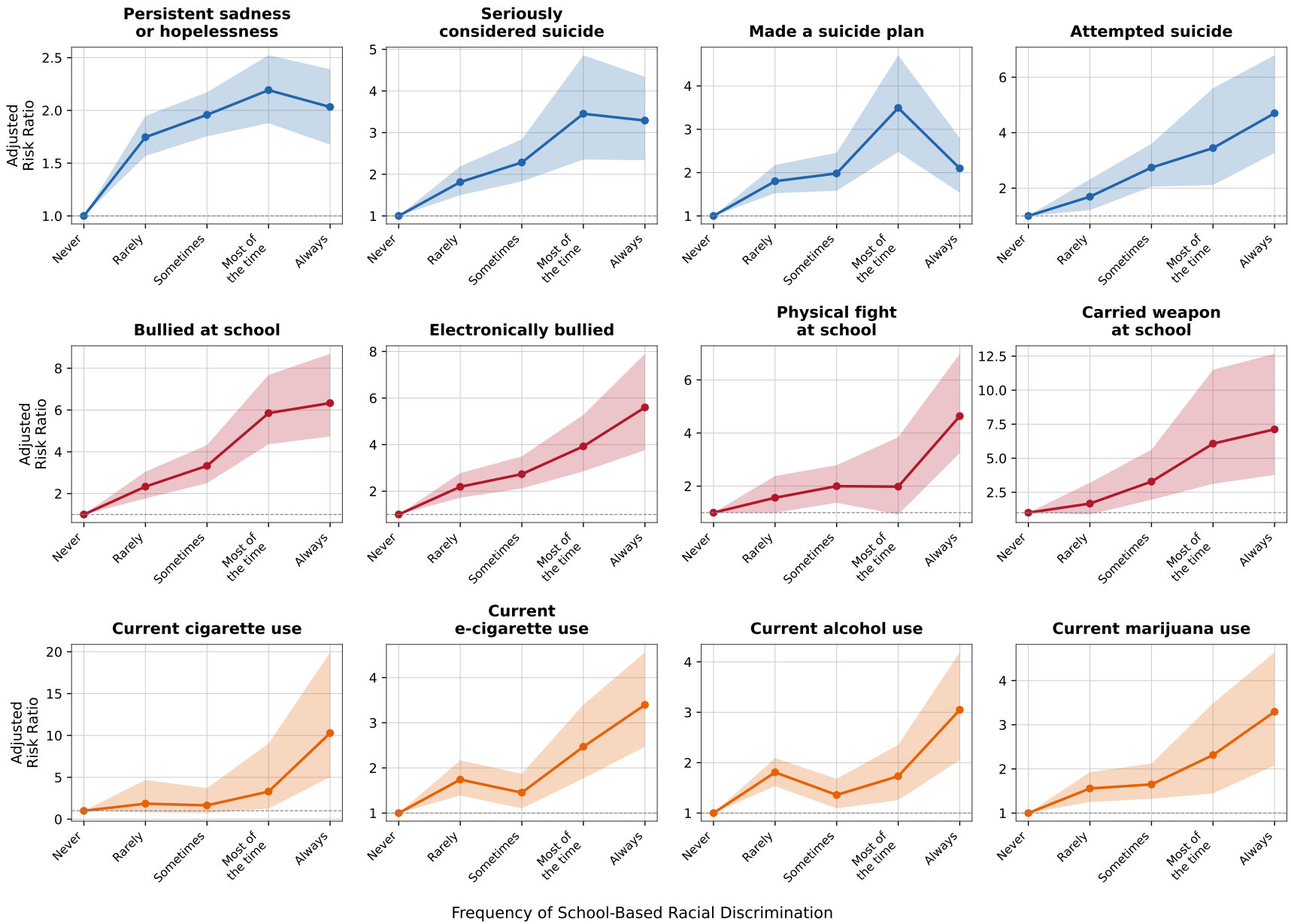
		Adjusted Risk Ratio (95% CI)				
Outcome	N	Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	821	1 [Ref]	1.64 (1.05–2.63)	1.09 (0.46–1.91)	1.81 (0.99–2.85)	1.84 (1.11–2.95)
Seriously considered suicide	817	1 [Ref]	2.16 (0.88–5.09)	2.15 (0.61–6.84)	6.31 (2.90–13.00)	1.46 (0.17–7.12)
Made a suicide plan	815	1 [Ref]	0.71 (0.22–2.16)	2.86 (0.98–6.86)	5.77 (2.57–11.22)	1.57 (0.22–6.32)
Attempted suicide	820	1 [Ref]	2.61 (0.87–7.40)	3.42 (0.56–15.58)	14.38 (5.60–34.57)	13.51 (4.01–35.59)
Violence						
Bullied at school	822	1 [Ref]	2.76 (0.88–7.82)	7.49 (2.40–19.39)	12.85 (4.57–31.68)	23.67 (10.26–51.61)
Electronically bullied	822	1 [Ref]	1.09 (0.28–3.38)	2.57 (0.81–7.05)	7.81 (4.06–13.57)	8.30 (4.45–16.41)
Physical fight at school	816	1 [Ref]	1.15 (0.44–2.73)	0.31 (0.05–2.09)	0.11 (0.01–2.37)	1.86 (0.11–19.87)
Carried weapon at school	823	1 [Ref]	1.76 (0.47–6.65)	2.37 (0.57–8.62)	2.28 (0.24–18.20)	0.00 (0.00–0.00)
Substance Use						
Current cigarette use	821	1 [Ref]	0.31 (0.06–1.74)	1.18 (0.12–9.17)	23.83 (5.57–68.14)	2.42 (0.18–21.38)
Current e-cigarette use	780	1 [Ref]	1.88 (0.70–4.34)	1.07 (0.34–2.96)	3.54 (1.78–6.26)	0.80 (0.08–3.95)
Current alcohol use	796	1 [Ref]	0.63 (0.17–1.97)	2.63 (0.99–6.82)	3.35 (0.75–8.31)	3.32 (0.47–9.63)
Current marijuana use	818	1 [Ref]	1.46 (0.50–3.43)	1.64 (0.62–3.56)	2.89 (1.29–5.32)	1.50 (0.04–5.33)

Native Hawaiian/Pacific Islander Students ($N = 92$)

		Adjusted Risk Ratio (95% CI)				
Outcome	N	Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	53	1 [Ref]	0.75 (0.16–5.21)	1.30 (0.09–8.35)	0.00 (0.00–0.00)	3.23 (1.34–13.71)
Seriously considered suicide	53	1 [Ref]	0.74 (0.02–20.75)	—	0.00 (0.00–0.00)	—
Made a suicide plan	53	1 [Ref]	—	—	0.00 (0.00–0.00)	—
Attempted suicide	53	1 [Ref]	—	—	0.00 (0.00–0.00)	—
Violence						
Bullied at school	53	1 [Ref]	—	0.00 (0.00–0.00)	—	—
Electronically bullied	53	1 [Ref]	—	—	—	—
Physical fight at school	53	1 [Ref]	—	—	—	—
Carried weapon at school	52	1 [Ref]	—	—	—	—
Substance Use						
Current cigarette use	53	1 [Ref]	—	—	—	—
Current e-cigarette use	51	1 [Ref]	0.00 (0.00–0.00)	—	0.00 (0.00–0.00)	—
Current alcohol use	51	1 [Ref]	—	—	0.00 (0.00–0.00)	—
Current marijuana use	53	1 [Ref]	0.00 (0.00–0.00)	—	0.00 (0.00–0.00)	—

Multiracial Students ($N = 4,296$)

		Adjusted Risk Ratio (95% CI)				
Outcome	N	Never	Rarely	Sometimes	Most of the time	Always
Mental Health						
Persistent sadness or hopelessness	2,738	1 [Ref]	1.73 (1.51–1.97)	1.87 (1.64–2.12)	1.88 (1.52–2.27)	2.11 (1.58–2.58)
Seriously considered suicide	2,736	1 [Ref]	1.81 (1.34–2.39)	2.21 (1.65–2.93)	3.18 (2.21–4.41)	3.90 (2.74–5.08)
Made a suicide plan	2,732	1 [Ref]	1.52 (1.09–2.15)	1.79 (1.34–2.33)	2.88 (1.91–4.07)	2.03 (1.19–3.31)
Attempted suicide	2,742	1 [Ref]	1.48 (1.01–2.27)	2.63 (1.78–3.81)	2.68 (1.27–5.13)	5.67 (3.57–8.31)
Violence						
Bullied at school	2,756	1 [Ref]	2.58 (2.06–3.30)	3.74 (2.93–4.72)	5.33 (3.97–6.96)	6.34 (4.67–8.25)
Electronically bullied	2,754	1 [Ref]	2.01 (1.46–2.81)	3.04 (2.24–4.14)	3.26 (2.02–5.34)	6.10 (3.60–9.18)
Physical fight at school	2,743	1 [Ref]	1.41 (0.78–2.41)	2.36 (1.31–4.01)	1.02 (0.34–3.02)	6.06 (3.42–11.07)
Carried weapon at school	2,759	1 [Ref]	1.76 (0.85–3.54)	3.40 (1.63–7.02)	3.55 (1.51–7.28)	7.85 (4.08–15.68)
Substance Use						
Current cigarette use	2,752	1 [Ref]	2.18 (0.97–4.94)	1.90 (0.70–4.77)	2.65 (0.54–11.12)	11.16 (5.05–23.91)
Current e-cigarette use	2,637	1 [Ref]	1.68 (1.20–2.32)	1.42 (1.02–1.90)	2.15 (1.46–3.19)	3.17 (1.98–4.61)
Current alcohol use	2,718	1 [Ref]	1.69 (1.41–2.02)	1.21 (0.83–1.74)	1.63 (1.10–2.30)	3.02 (1.86–4.34)
Current marijuana use	2,729	1 [Ref]	1.52 (1.15–2.02)	1.69 (1.31–2.23)	2.21 (1.25–3.50)	3.56 (2.13–5.15)



eFigure 1. Adjusted risk ratios for health outcomes by frequency of school-based racism among non-White US high school students, 2023 YRBS ($N = 9,312$). Risk ratios from logistic regression computed via simulation from the variance-covariance matrix (King et al, 2000), adjusting for sex, age, race/ethnicity, English language proficiency, and grades. Shaded bands indicate 95% confidence intervals. Reference category is Never.

Data and Code Availability. Data are publicly available from the Centers for Disease Control and Prevention.[4] Code to download the data and reproduce all analyses is available at `https://github.com/tlcaputi/yrbs-racism`.